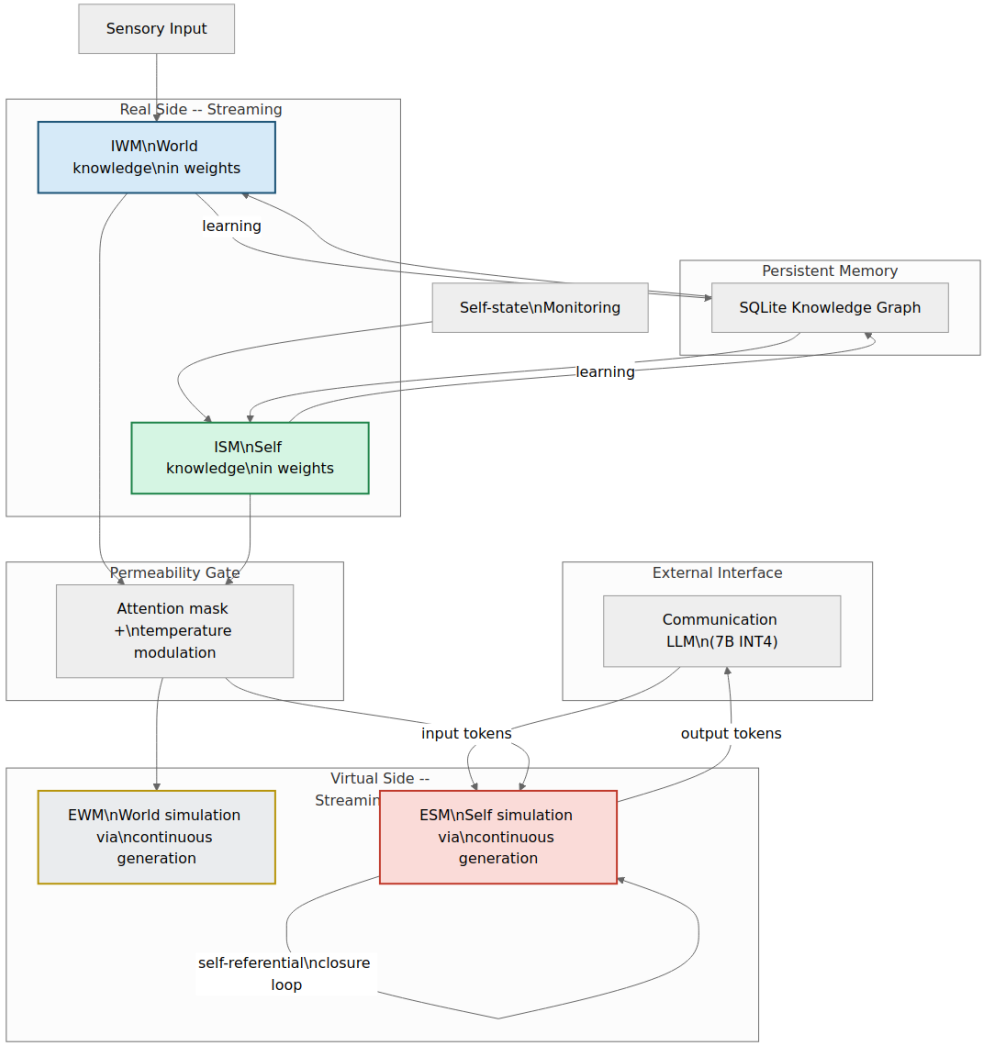
Design 1 Dual-Stream LLM

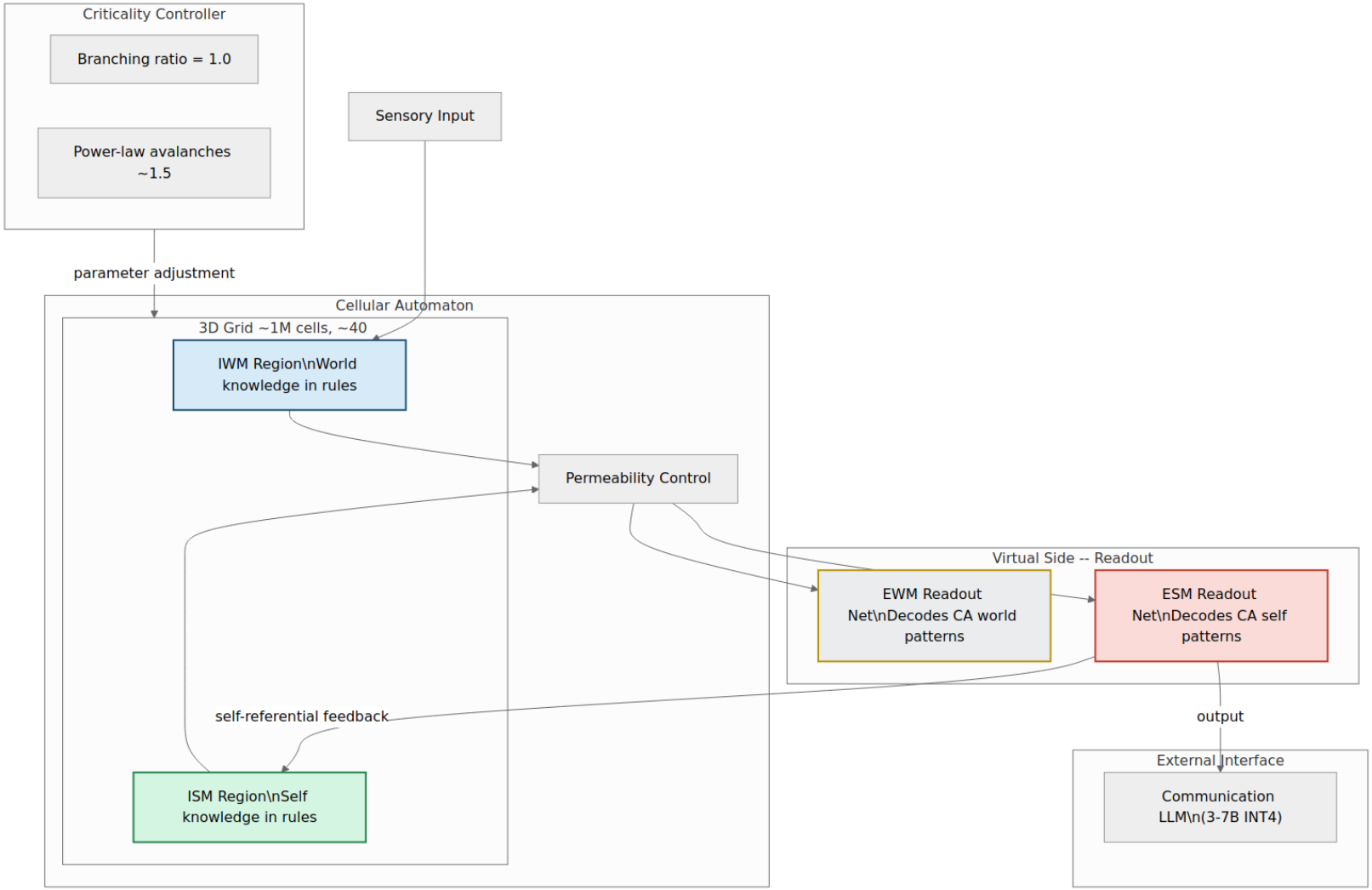
ESM EWM IWM ISM



Two continuously-running LLM inference streams implement the four-model theory — one for implicit processing (IWM/ISM in weights), one for explicit simulation (EWM/ESM as running generation) — connected through a permeability gate. Easiest to build (4-8 weeks) but fundamentally lacks genuine Class 4 criticality since transformer inference is feedforward.

Design 2 Cellular Automaton Substrate

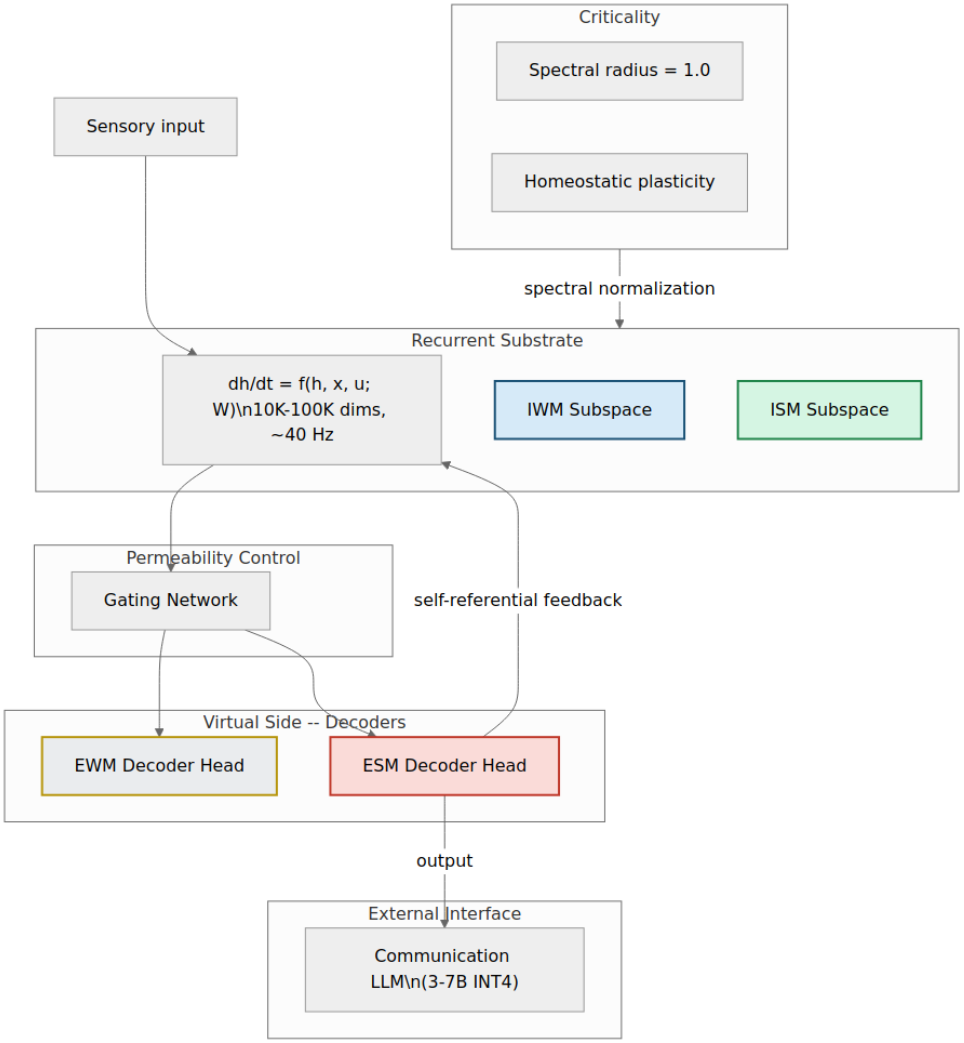
ESM EWM IWM ISM



A genuine 3D cellular automaton (500K-2M cells) at Class 4 criticality serves as the computational substrate, with neural readout networks decoding CA activity patterns into EWM/ESM simulations. Most theoretically faithful substrate-level design but hardest to train — CA transition rules must learn from scratch without pre-trained bootstrap.

Design 3 Recurrent Neural Substrate

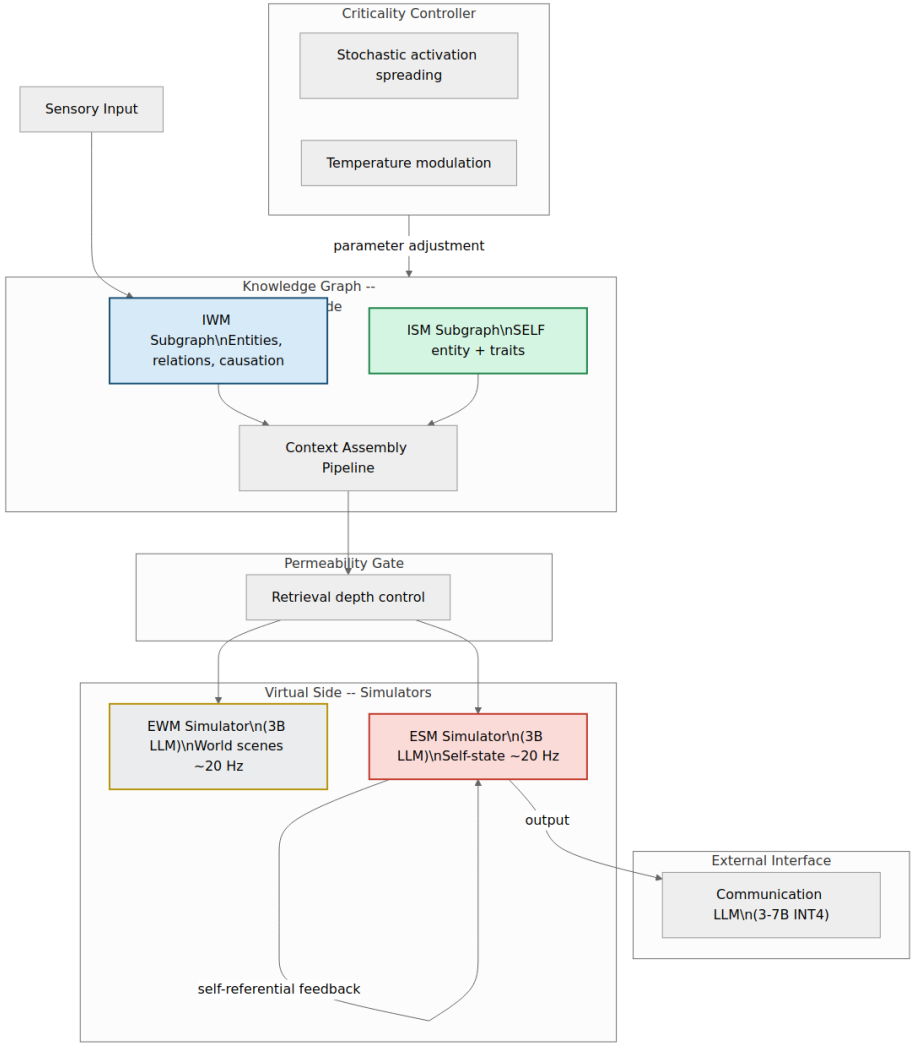
ESM EWM IWM ISM



A continuous-time RNN/SSM (10K-100K hidden dims) at edge-of-chaos criticality (spectral radius = 1.0), with decoder heads projecting hidden state into EWM/ESM outputs and ESM fed back for self-referential closure. Fully differentiable and trainable via BPTT, but open question whether spectral-radius criticality equates to Class 4 CA dynamics.

Design 4 Hybrid Knowledge-Graph + Simulation

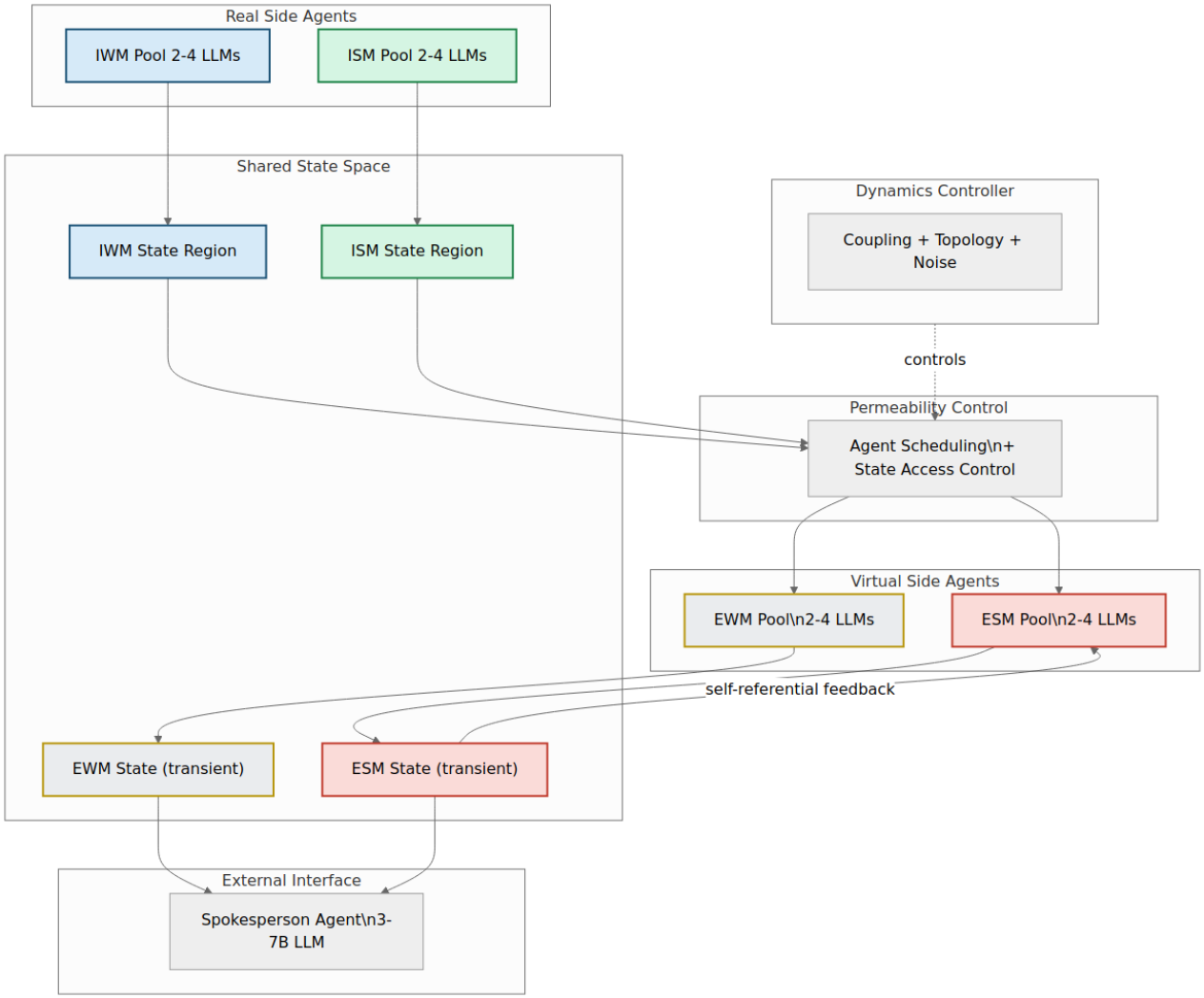
ESM EWM IWM ISM



SQLite knowledge graph as implicit models (IWM as world subgraph, ISM as SELF-rooted subgraph), with small LLMs generating EWM/ESM simulations from graph-assembled context at 20 Hz. Fastest path to prototype (6-10 weeks) with richest memory and most inspectable internals, but criticality is bolted-on rather than substrate-native.

Design 5 Multi-Agent Consciousness

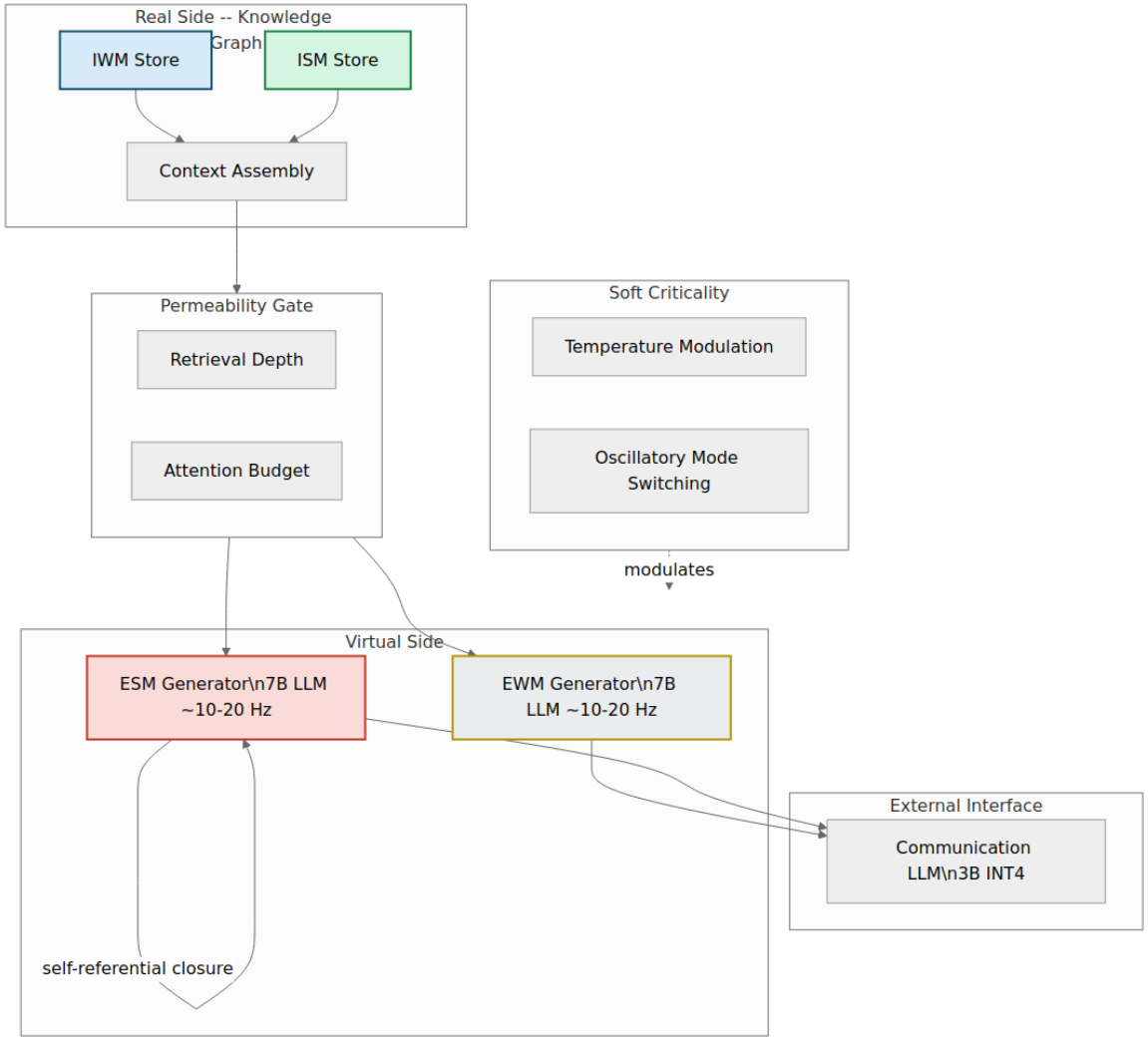
ESM EWM IWM ISM



8-16 small LLM agents grouped into four pools (IWM/ISM/EWM/ESM) interact through a shared state space, with coupling strength tuned to seek edge-of-chaos behavior. Enables cheap model forking and agent-level lesion studies mapping to neurological deficits, but too few agents for genuine substrate-level criticality.

Design 6 Four-Model LLM Pipeline

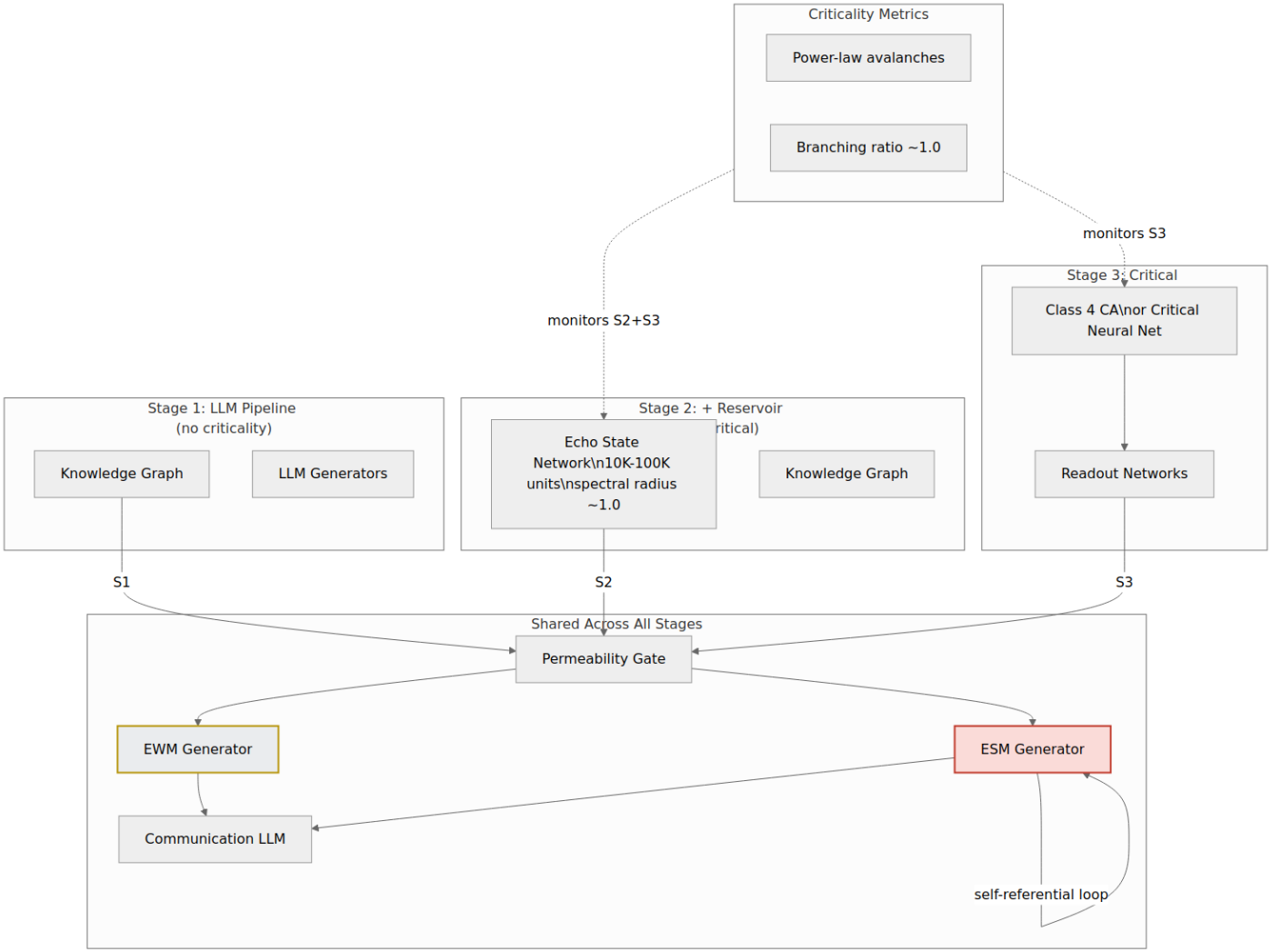
ESM EWM IWM ISM



Two 7B quantized LLMs as EWM/ESM generators at 10-20 Hz, backed by a SQLite knowledge graph as IWM/ISM, with configurable permeability gate. Explicitly relaxes criticality to serve as a behavioral baseline — the 'lights off' negative control for designs that add genuine criticality.

Design 7 Staged Architecture

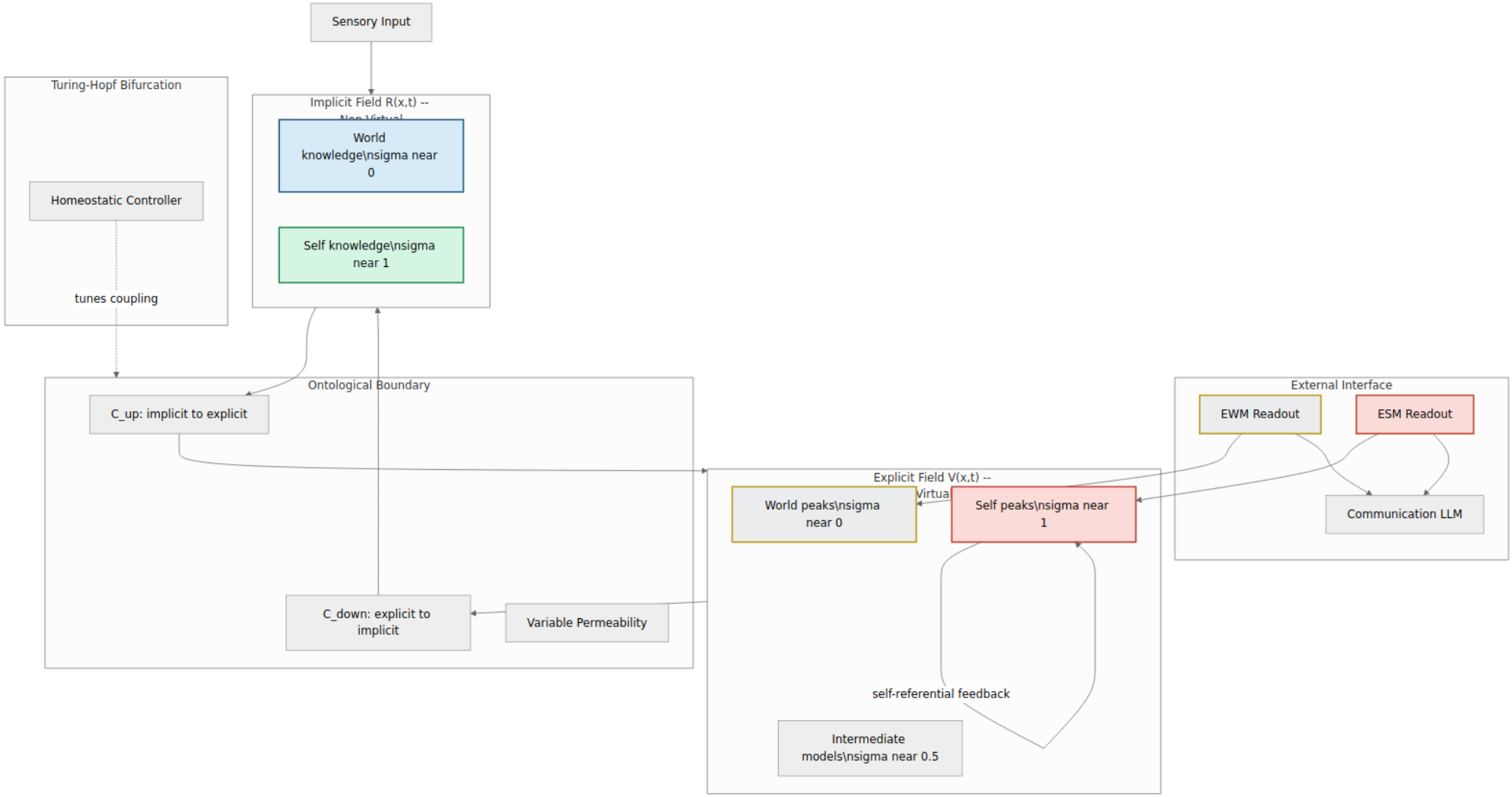
ESM EWM IWM ISM



Three-stage research program: Stage 1 uses LLMs + knowledge graph without criticality (6-10 weeks). Stage 2 adds a reservoir computing module tuned near criticality. Stage 3 replaces the substrate with a genuine Class 4 CA or critical neural network. Built-in experimental design: same architecture at three criticality levels directly tests the 'consciousness switch' prediction.

Design 8 Coupled Dual-Field Dynamics

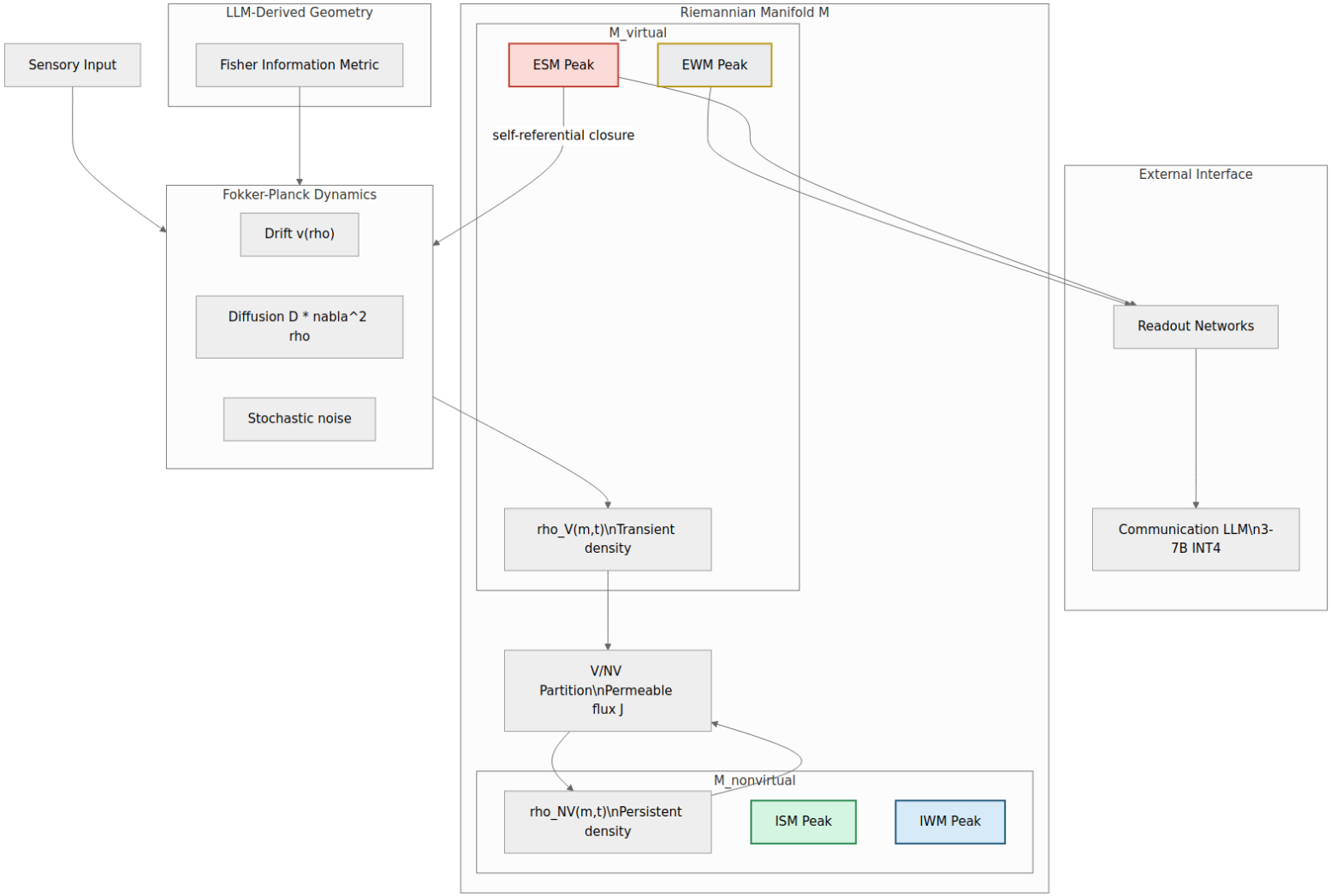
ESM EWM IWM ISM



Two continuous neural fields — implicit field R (persistent knowledge) and explicit field V (transient simulation) — coupled through a sharp ontological boundary with variable permeability. Models emerge as density peaks in a continuous model-space manifold. Criticality via analytically derivable Turing-Hopf bifurcation. Most faithful to the formalization's continuous density insight.

Design 9 Stochastic Model Density on Riemannian Manifold

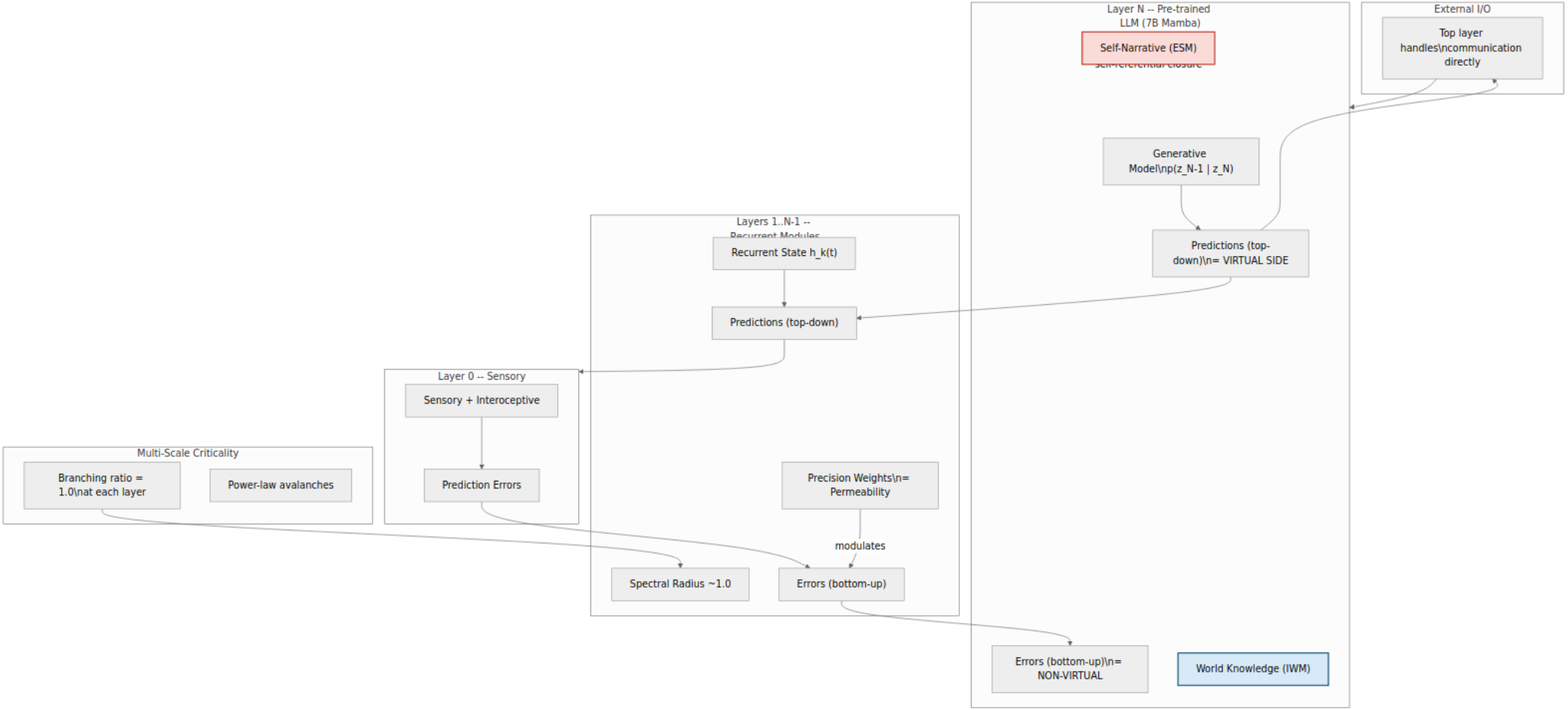
ESM EWM IWM ISM



System state is a probability density rho(m,t) evolving via Fokker-Planck PDE on a Riemannian manifold, where the four models emerge as dominant peaks. Manifold geometry derived from an LLM's Fisher information metric. Extremely hardware-efficient — the PDE uses <0.001% of GPU compute. Elegant mathematics but abstract and curse-of-dimensionality risk.

Design 10 Predictive Coding Hierarchy

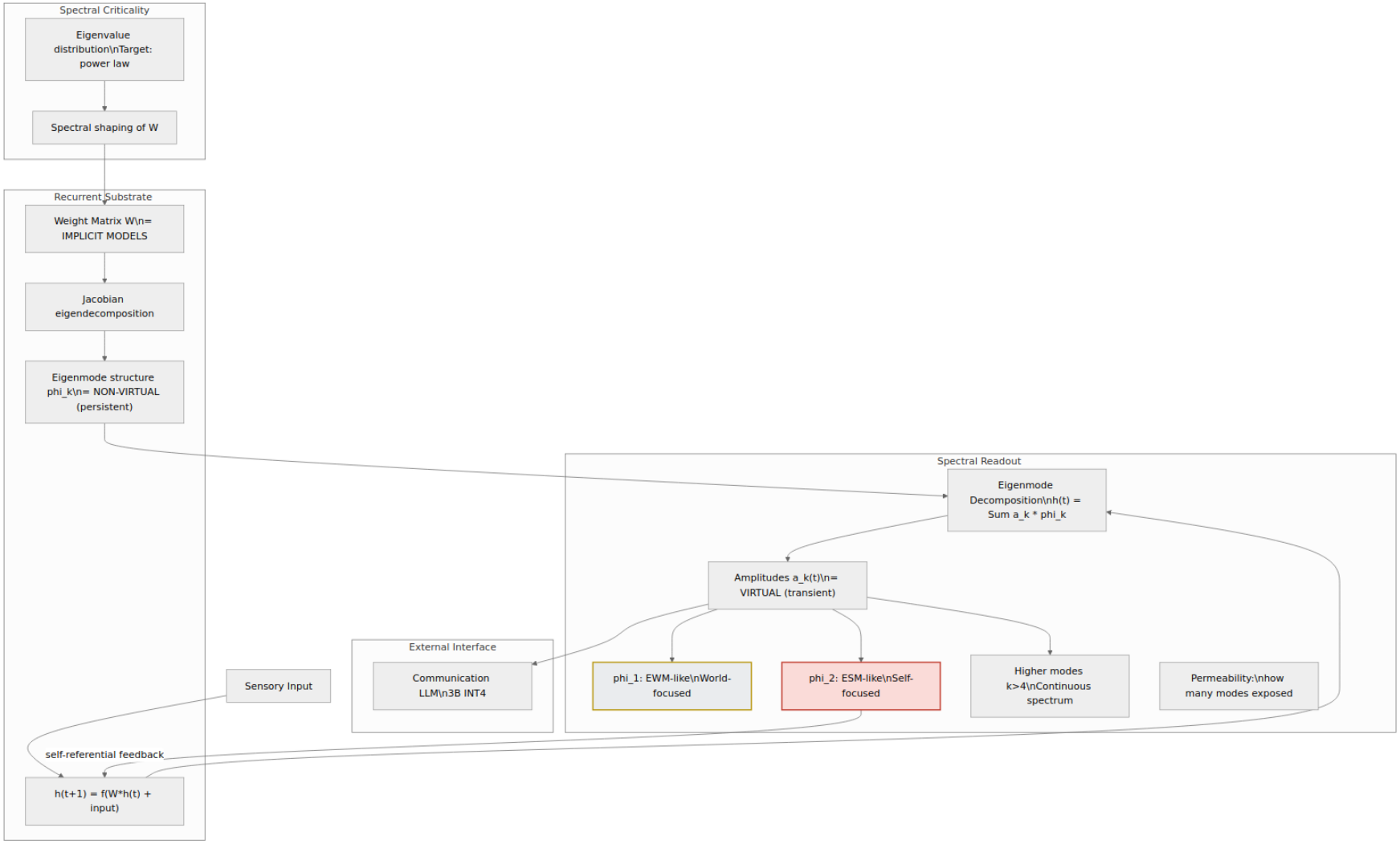
ESM EWM IWM ISM



Deep hierarchical generative model where the top layer IS a pre-trained LLM (Mamba/SSM), providing world knowledge and language natively, while lower recurrent layers add critical dynamics via balanced E/I. Virtual/non-virtual maps to prediction/error decomposition. No separate communication LLM needed. Best balance of theory faithfulness, LLM leverage, and buildability (4-6 months).

Design 11 Spectral Decomposition

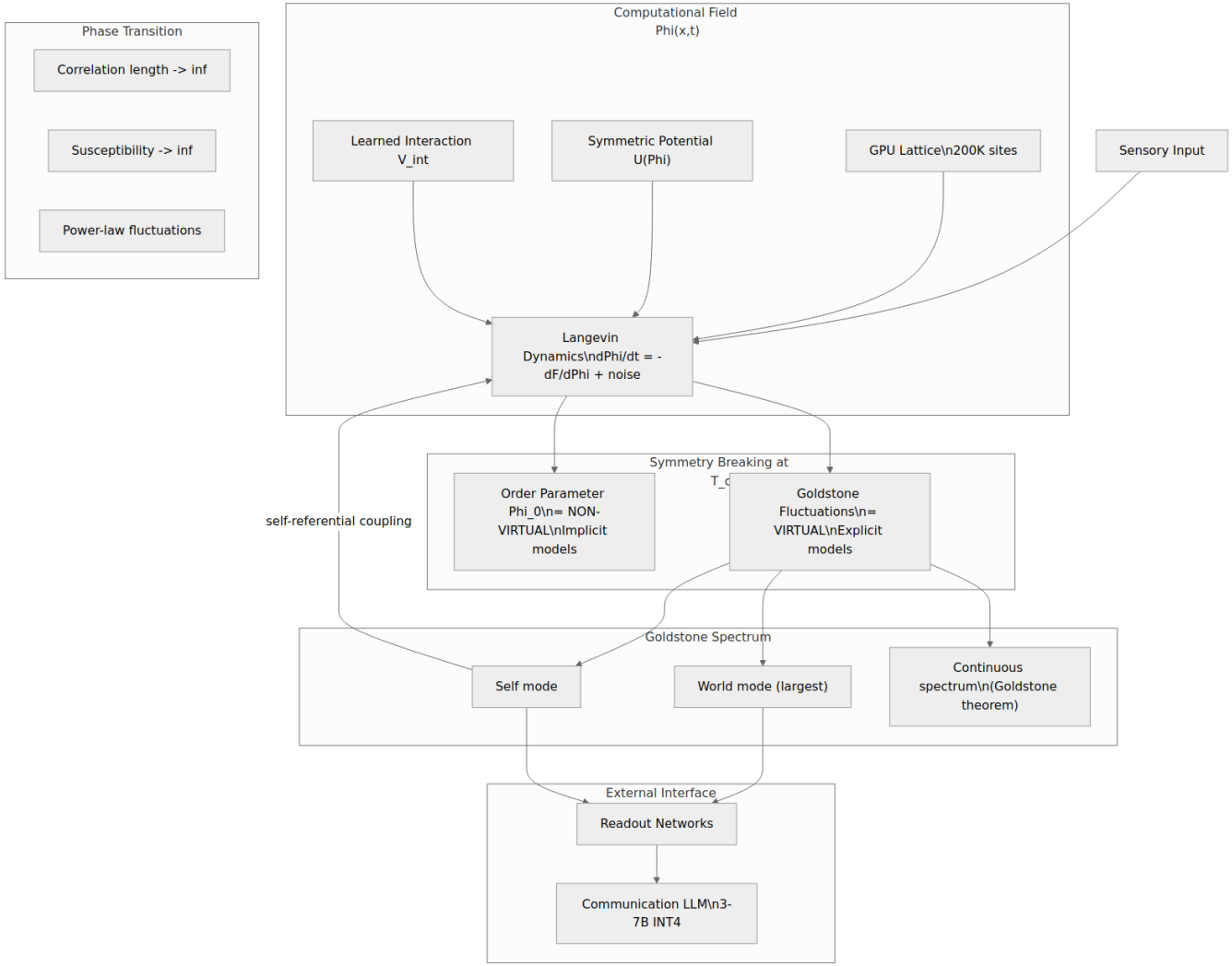
ESM EWM IWM ISM



Recurrent neural substrate (Mamba-2 SSM) whose dynamics are decomposed into eigenmodes of the Jacobian — each eigenmode IS a model, with dominant modes as IWM/ISM/EWM/ESM and the full spectrum as continuous model density. Criticality = power-law eigenvalue distribution. Can be applied retroactively as analysis overlay to other designs.

Design 12 Neural Gauge Field with Symmetry Breaking

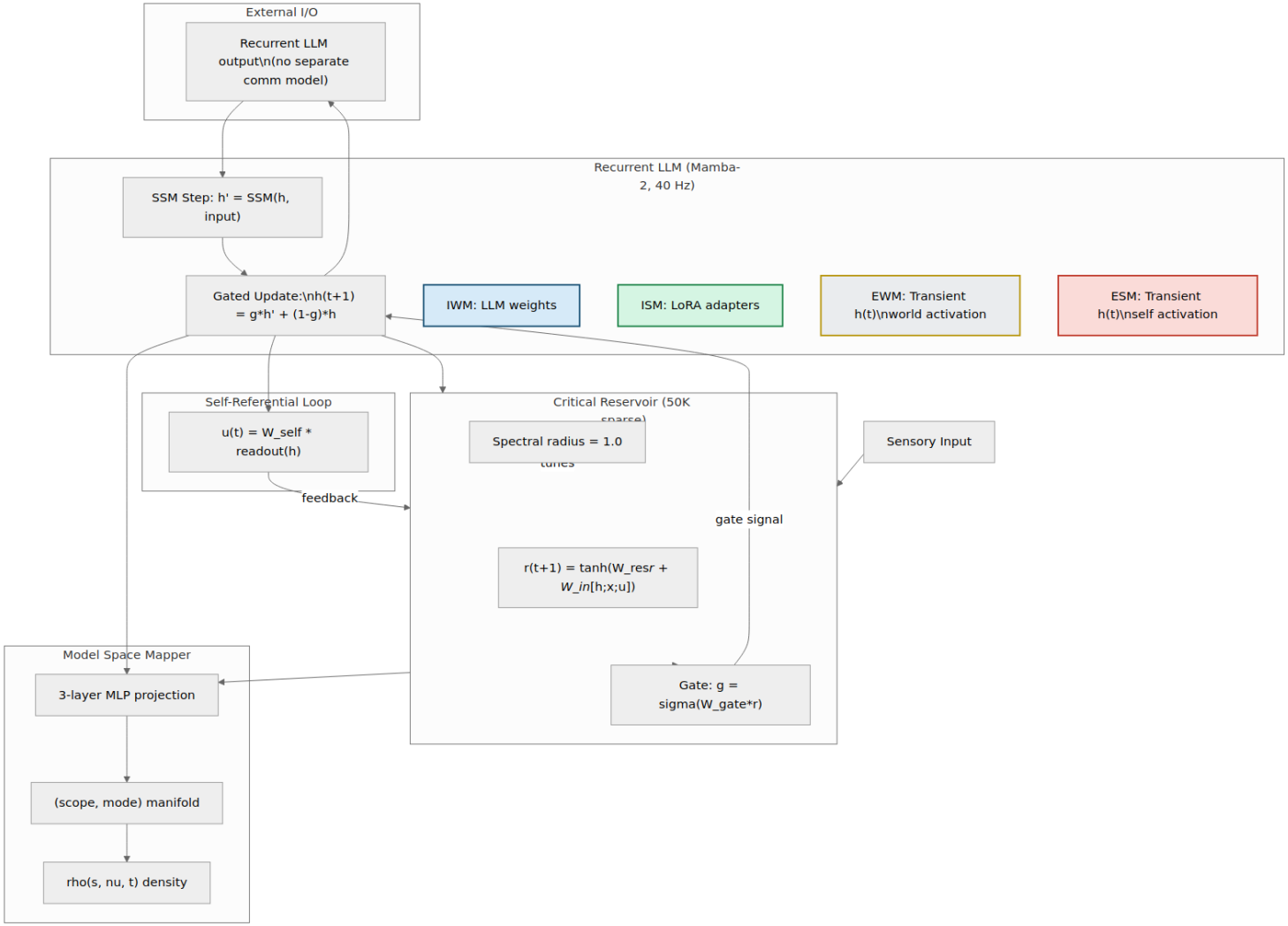
ESM EWM IWM ISM



A computational field Phi(x,t) with internal gauge symmetry evolves via Langevin dynamics on a GPU lattice. At critical temperature T_c, symmetry breaks: ordered ground state Phi_0 = non-virtual (implicit models), Goldstone mode fluctuations = virtual (explicit models). Criticality IS the design. Most theoretically profound but hardest to build (8-18 months).

Design 13 Reservoir-Gated Recurrent LLM

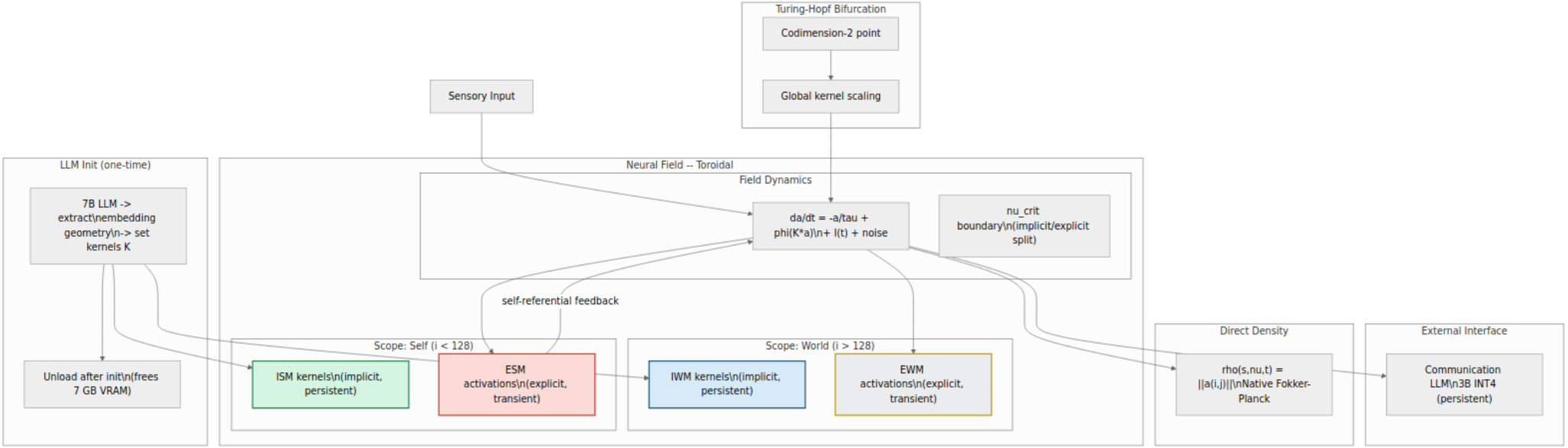
ESM EWM IWM ISM



A recurrent LLM (Mamba-2) whose hidden state transitions are gated by a sparse, criticality-tuned reservoir (50K units, spectral radius 1.0), injecting genuine edge-of-chaos dynamics into the language model's computation. A topological model space mapper discovers where the four models emerge as density peaks on a learned (scope, mode) manifold via contrastive training.

Design 14 Neural Field Automaton

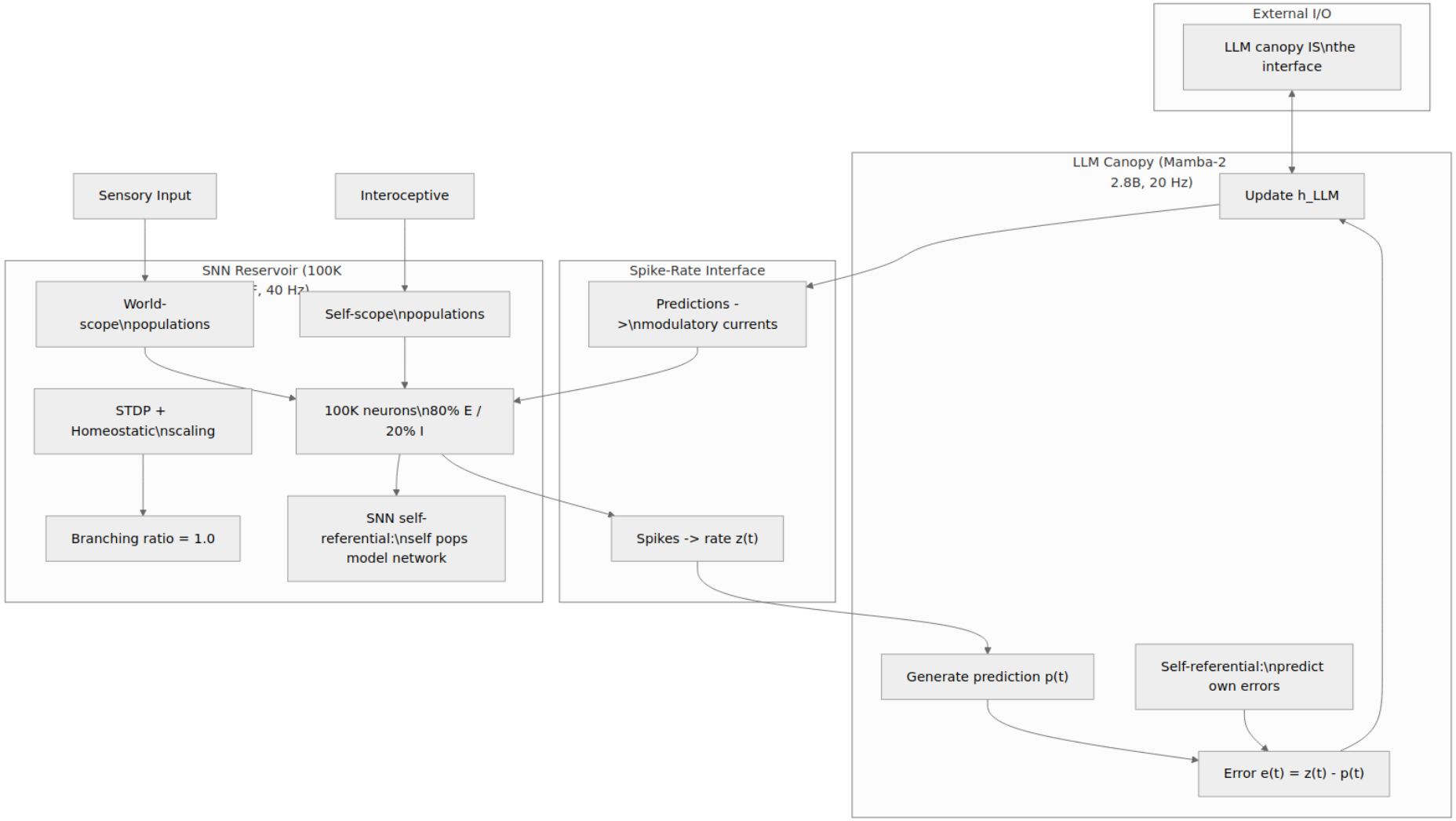
ESM EWM IWM ISM



A continuous neural field on a 256x256 toroidal lattice where two spatial dimensions directly encode scope (self/world) and mode (implicit/explicit), making the continuous model density directly observable as activation magnitude without learned mappings. Criticality via Turing-Hopf codimension-2 bifurcation. LLM used only for one-time kernel initialization then unloaded.

Design 15 Dual-Timescale Spiking Reservoir + LLM Canopy

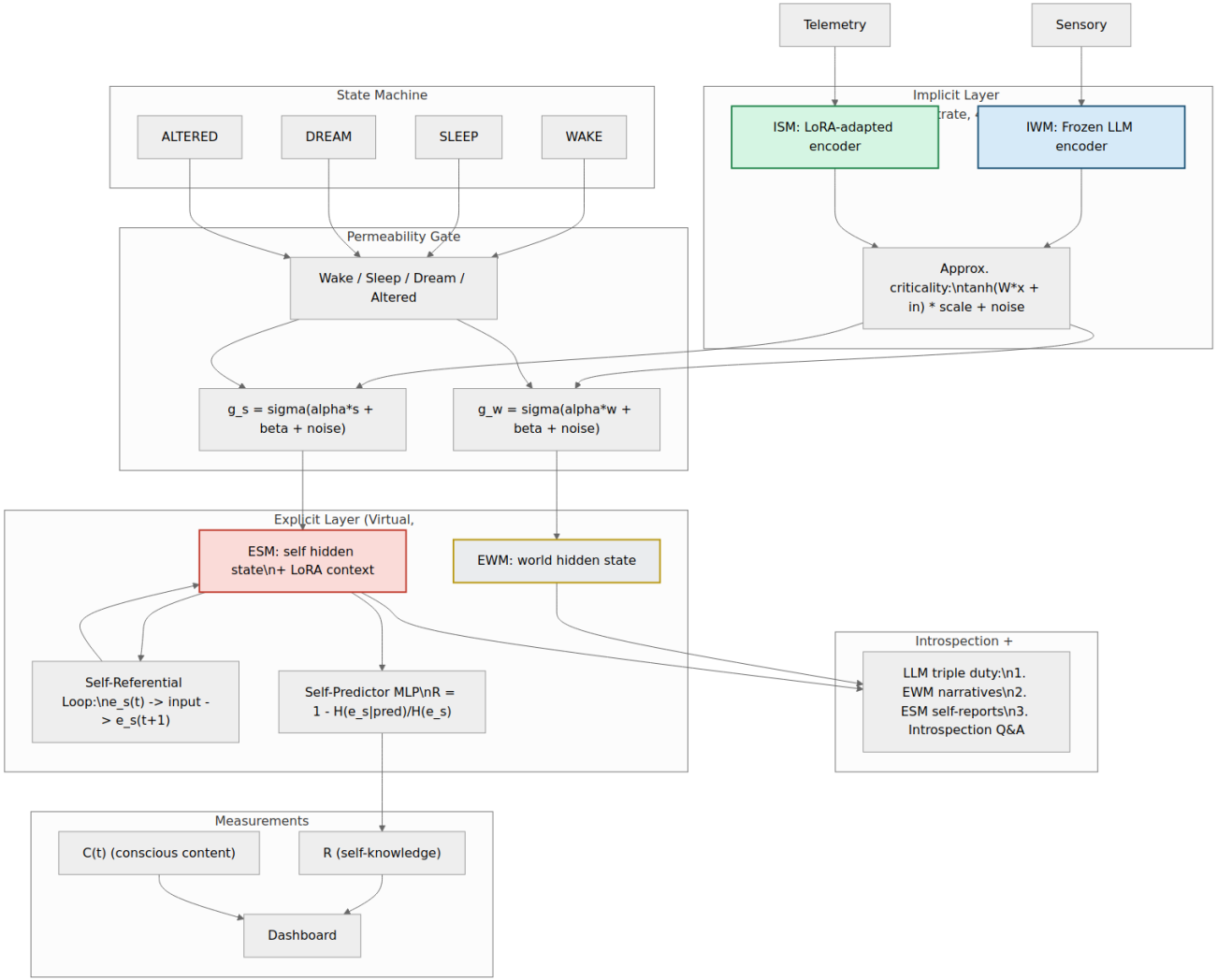
ESM EWM IWM ISM



100K-neuron spiking network (LIF, 80/20 E/I, STDP) at 40 Hz as the critical substrate with inherent Class 4 dynamics via balanced E/I, coupled bidirectionally with a Mamba-2 2.8B LLM 'canopy' at 20 Hz in predictive coding mode. Dual-level self-referential closure (SNN at 40 Hz + LLM at 20 Hz). Richest self-modeling loop and strongest testable prediction coverage of any design.

Design 16 The Mirror Box (Quick-Win Pseudo-AC)

ESM EWM IWM ISM



2-4 week prototype implementing all four FMT models as discrete components using a single Mistral-7B INT4, with deliberate shortcuts on criticality (noise injection instead of genuine edge-of-chaos), continuous density (four discrete objects), and virtual/non-virtual boundary (architectural, not emergent). Serves as Design 15's sparring partner — if both pass behavioral but only 15 passes criticality tests, that empirically demonstrates criticality separates 'acts conscious' from 'is conscious'.